



**INSTITUT
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DE PARIS**

Nested Investor Syndication Networks

A Network Science Approach to Venture Capital Ecosystems

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Outline

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- 2 Understanding Bipartite Networks
- 3 Nestedness: From Ecology to Finance
- 4 Data & Methodology
- 5 Key Results
- 6 Discussion & Implications
- 7 Conclusion

The Role of Syndication in Innovation Financing

Syndication = investors jointly funding the same venture

- **Role:** risk-sharing and quality signaling
- **Effects:** complex coordination patterns
- **Correlated to:** information asymmetries in investment markets

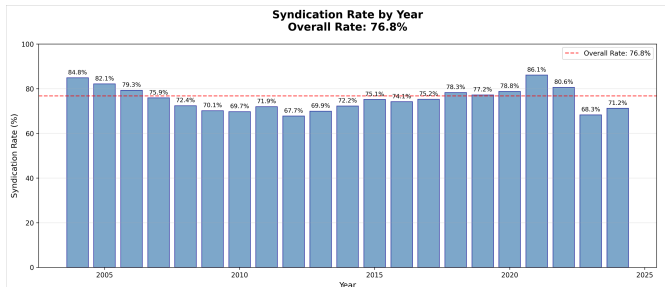


Figure: Syndication trends in US market

Research Gap & Motivation

Current Limitation

Traditional approaches models networks links regardless investor intrinsic characteristics.

Our Approach

Apply Network Science theories to analyze coordination patterns in **early-late stage investor syndication networks** in the U.S. market.

- Broader understanding of systemic coordination patterns
- Need for **multidisciplinary perspective**:
 - Finance & Economic Sociology
 - Network Science & Ecology

Why Early-Late Stage Distinction Matters

Early-Stage Investors

- Focus on **seed & Series A**
- Higher risk tolerance
- Hands-on mentoring
- Smaller investment amounts
- Technology & market validation

Late-Stage Investors

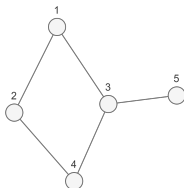
- Focus on **Series B+**
- Growth capital orientation
- Scaling expertise
- Larger investment amounts
- Market expansion & profitability

Key Insight

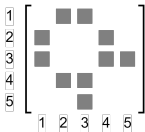
These different roles justify modeling them as **separate node types** in a bipartite network!

Unipartite vs. Bipartite Network Representations

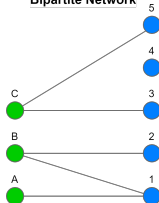
Unipartite Network



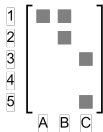
Adjacency Matrix



Bipartite Network

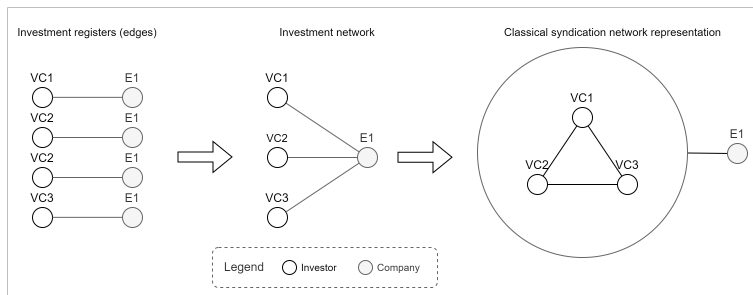


Biadjacency Matrix



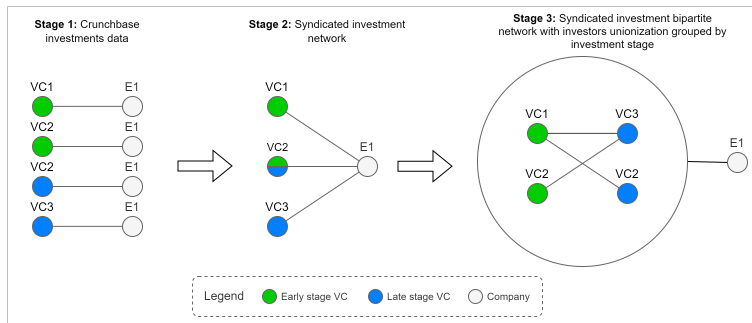
- **Unipartite:** Single node set, square adjacency matrix
- **Bipartite:** Two disjoint node sets, rectangular biadjacency matrix
- Bipartite structure **essential** for detecting nestedness patterns

From Classical to Bipartite Syndication Models



- Connect all co-investors equally
- Obscures role differentiation and hierarchical patterns

Our Early-Late Stage Bipartite Model



- Preserves sequential flow of capital across funding stages
- Enables analysis of hierarchical organization patterns

What is Nestedness?

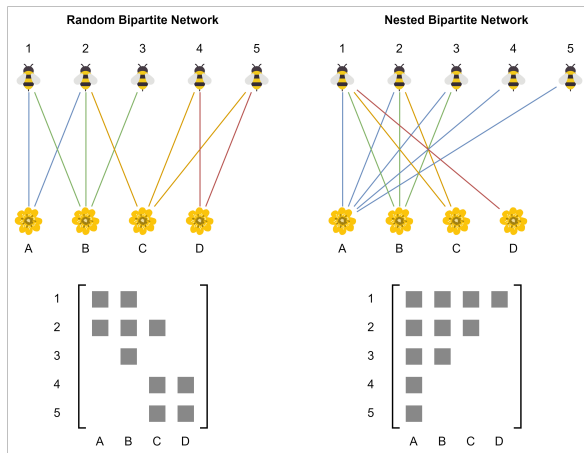


Figure: Nestedness in pollinator-plant networks. **Specialists** interact with subsets of the same partners as **generalists**

Why Nestedness Matters for Innovation

Implications for VCs:

- Information flow from generalists to specialists
- Hierarchical access to opportunities
- Efficiency vs. accessibility trade-offs

Implications for Startups:

- Network position affects funding access
- "Gatekeeper" investors control broader networks
- Geographic clustering effects

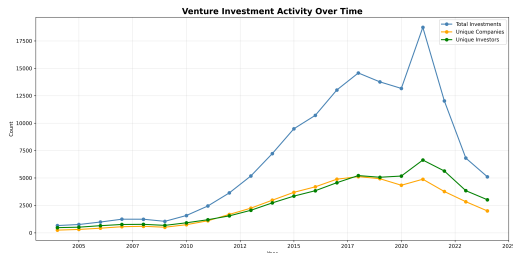
Research Questions

Do VC networks exhibit nestedness? How does geography influence it? When does it emerge over time?

Dataset Overview

After Cleaning:

- 17 k companies
- 11 k investors
- 100 k records
- **\$3.2 tri** volume



Geographic & Temporal Scope:

- Focus: U.S. companies (international investors included)
- Period: 2004-2024
- Minimum funding threshold: \$150,000

① Stage Classification:

- Early stages: Angel, Pre-seed, Seed, Series A
- Late stages: Series B through Series I

② Bipartite Network Creation:

- Two disjoint investor sets: Early vs. Late
- Edges = co-investment in same companies
- 169,679 investment pairs across 3,666 startups

③ Community Detection:

- Greedy modularity optimization
- Identifies 168 communities
- Focus on 7 communities with 150+ investors

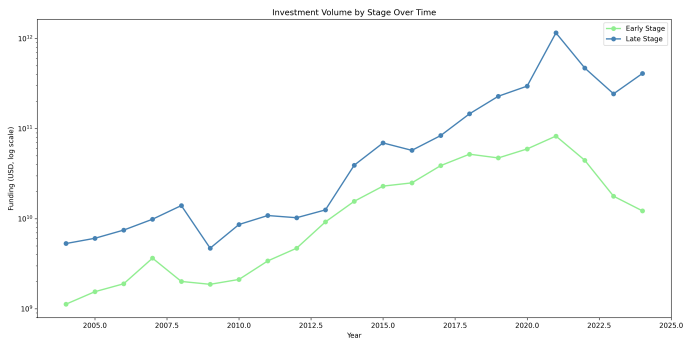
Analytical Framework

Network Analysis:

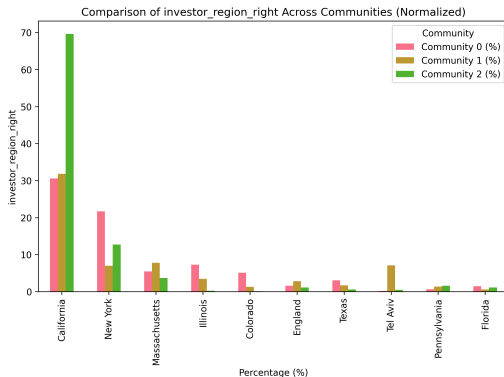
- Degree distribution
- Geographic clustering
- Investment characteristics

Nestedness Analysis:

- NODF metric calculation
- Curveball null models (1,000)
- Temporal evolution analysis



Geographic Clustering: Silicon Valley Dominance



- **Community 0:** Balanced U.S. distribution
- **Community 1:** New York + internationally diversified
- **Community 2:** Silicon Valley concentration

Community Structure: Heterogeneous Organization

Community	Investors	Investment Share
0	4,248	19.4%
1	4,089	10.4%
2	3,959	33.6%
Cross-community	—	35.1%

Table: Distribution of co-investment activity

- **Community 2:** Despite similar size, handles 75% more investments than Community 0
- Substantial cross-community collaboration (35.1%)

Main Finding: Significant Nestedness in Silicon Valley

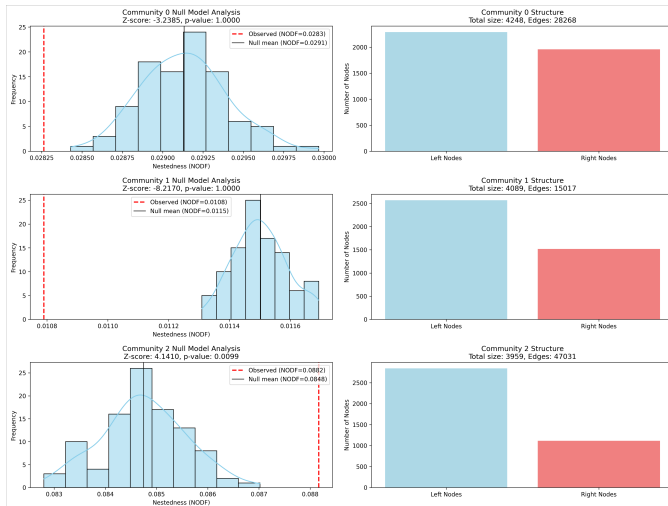
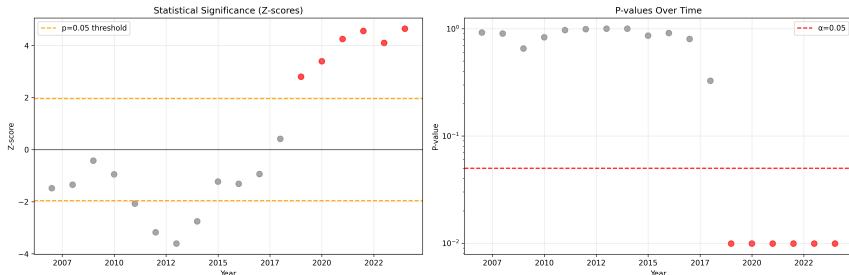


Figure: Observed vs. null model nestedness distributions

Temporal Evolution: Progressive Development

Nestedness Evolution Analysis - Community 2



- **2007-2018:** Significantly low nestedness
- **2012+:** Gradual increase begins
- **2019:** Statistical significance achieved
- **2019-2024:** Sustained high nestedness

Why Silicon Valley? Geographic Clustering Effects

Silicon Valley advantages:

- Dense information networks
- Proximity to talent & infrastructure
- Shared risk assessment practices
- Established trust relationships

Comparison with Comm. 0:

- Similar size & U.S. focus
- But **75% lower** transaction volume
- Non-nested organization
- Less efficient capital allocation

Key Insight

Network **topology**, not just size or geography, determines ecosystem efficiency.

Robustness Trade-offs: Stability vs. Vulnerability

Ecological Analogy:

- Nested networks are **resilient** to random failures
- But **vulnerable** to targeted hub removal
- "Mutualistic trade-offs" in ecosystem stability

VC Implications:

- Stable against random investor departures
- Fragile to key hub investor exits
- Concentration risk in Silicon Valley

Broader Applications

For Policymakers

- Support networking platforms and regional clusters
- Monitor concentration risks from hub dominance
- Foster inclusive access to innovation ecosystems

For Investors and Companies

- Strategic network positioning and partnership selection
- Understanding funding ecosystem structure for better access
- Evidence-based approaches to fostering funding trends

Three Key Contributions

1. **Network topology matters more than size:** Nestedness organization drives investment efficiency (not just community size or geographic distribution alone)
2. **Bridging ecology and finance:** Successfully applied ecological concepts (nestedness, bipartite networks) to understand venture capital syndication patterns (novel interdisciplinary approach)
3. **Geographic clustering enables hierarchy:** Silicon Valley's exceptional concentration facilitates nested organization

Limitations & Future Research

Limitations

- Crunchbase data may under-represent certain sectors/regions
- Focus on U.S. companies limits international generalizability
- Cannot establish causality between network structure and outcomes

Future Research Directions

- Causal analysis of network evolution mechanisms
- Cross-country comparison of investor network structures
- Connectance thresholds for nestedness emergence
- Dynamic modeling of network rewiring processes
- Investor-level nestedness contribution analysis

Questions & Discussion

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